



**Informatique et systèmes
de communication ISC**



Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

Signal and information (SigInf)

204.1

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Lesson 6 – part 1. Medical Image processing and Real- world data

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1. MRI

Magnetic Resonance Imaging

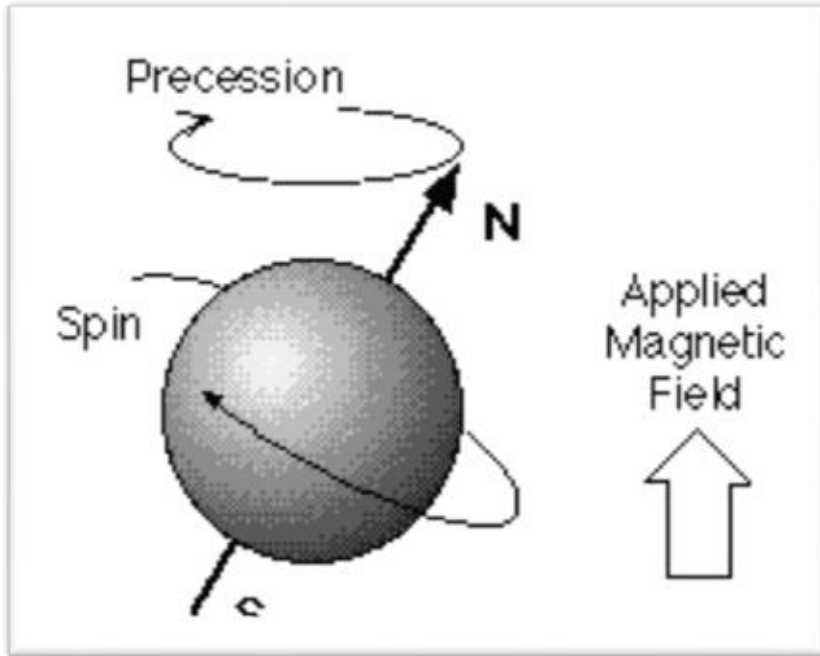


Siemens Healthineers Prismafit, 3T, Erlangen, GE

MRI is a non-invasive medical imaging technique which provides detailed anatomical (structural), functional and metabolic images of organs;

It is often used for disease detection, diagnosis, and treatment monitoring.

Magnetic Resonance Imaging



MRI is based on sophisticated technology that excites and detects the change in the direction of the rotational axis of protons found in the water that makes up living tissues.

1. Protons have a positive charge and a spin;
2. They behave like little magnets, i.e. they induce a magnetic field.

Magnetic Resonance Imaging

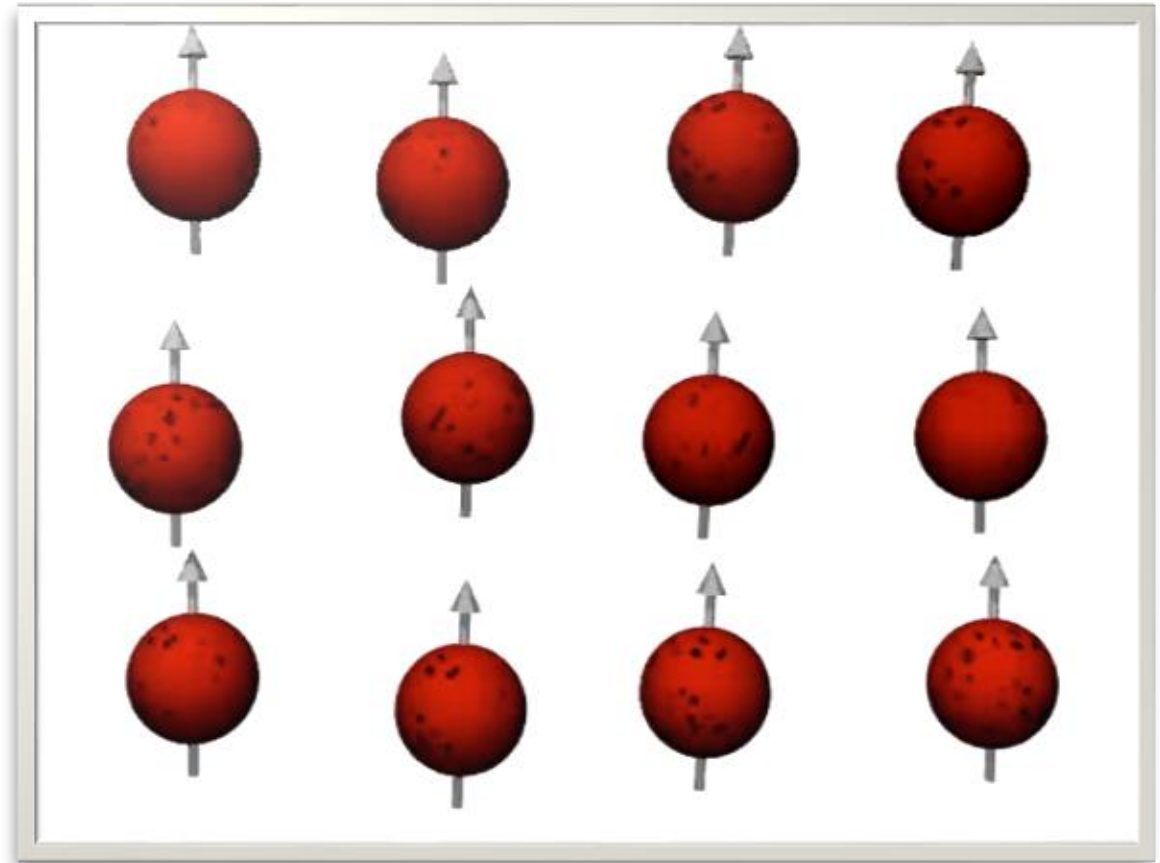
- ▶ Once immersed in a stronger magnetic field (B_0), protons align.
- ▶ The magnetic moment of the nuclei carries relevant information concerning the tissue.

First observation: we can't see them while they keep being parallel to B_0 .

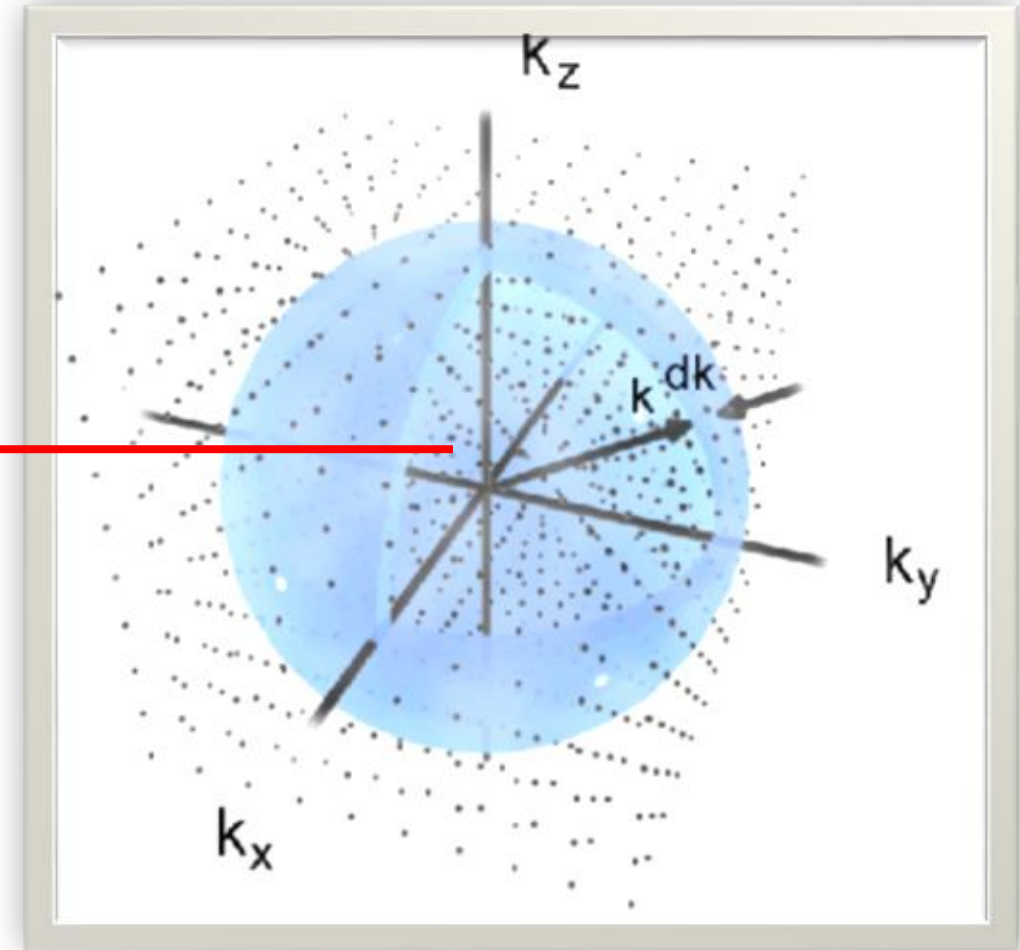
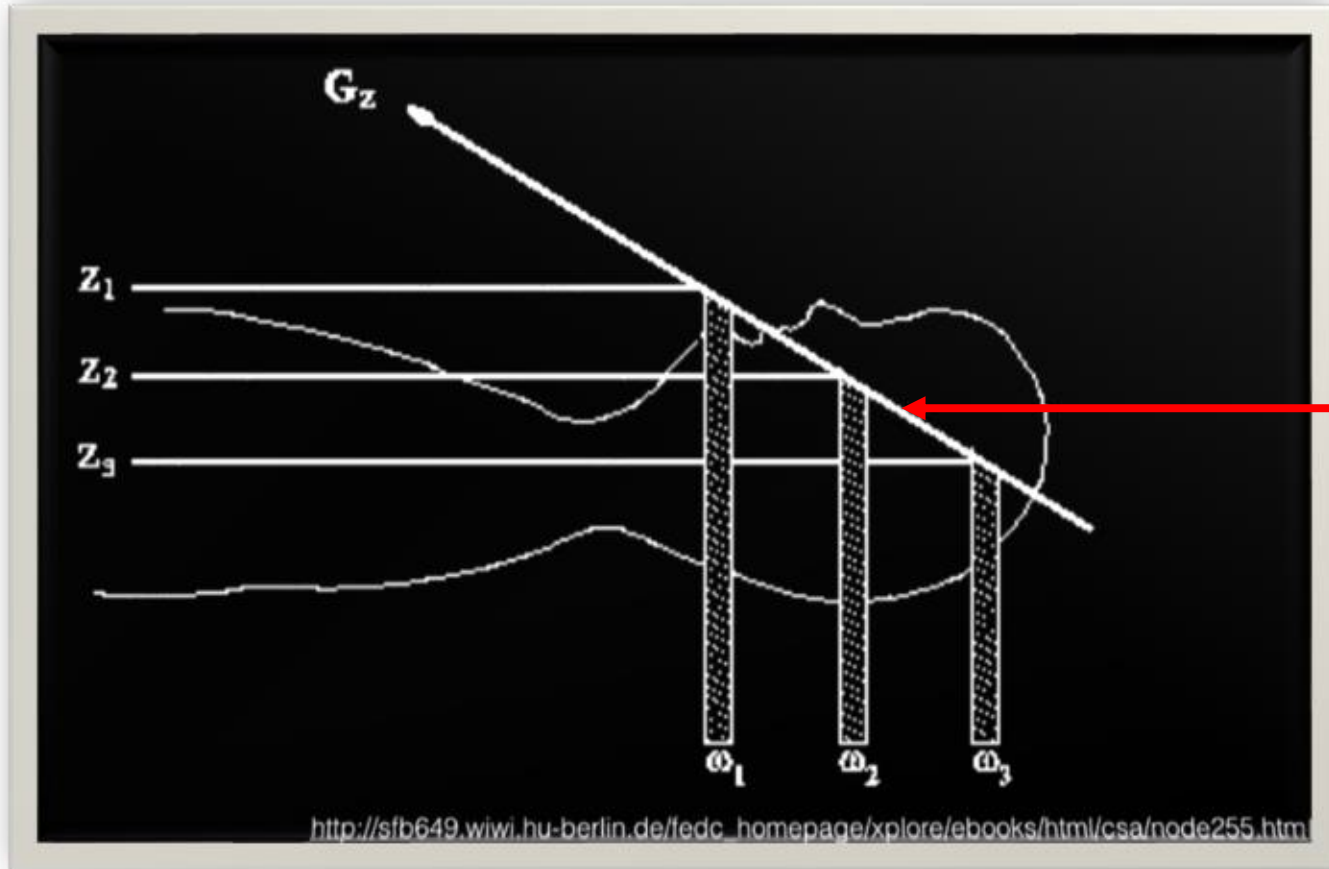
Due to this, they have a magnetic field and can be seen as little bar magnets.

Magnetic Resonance Imaging

- ▶ Inject energy in the system with a Radio-Frequency to bend the polarization of the nuclei, in a way that resonant nuclei will return to parallel magnetization together (i.e. encoded at the same frequency).
- ▶ Different types of pulses give different contrasts. Our contrast depends on:
 - ▶ T1 values
 - ▶ T2 values
 - ▶ Proton density



Magnetic Resonance Imaging



Magnetic Resonance Imaging

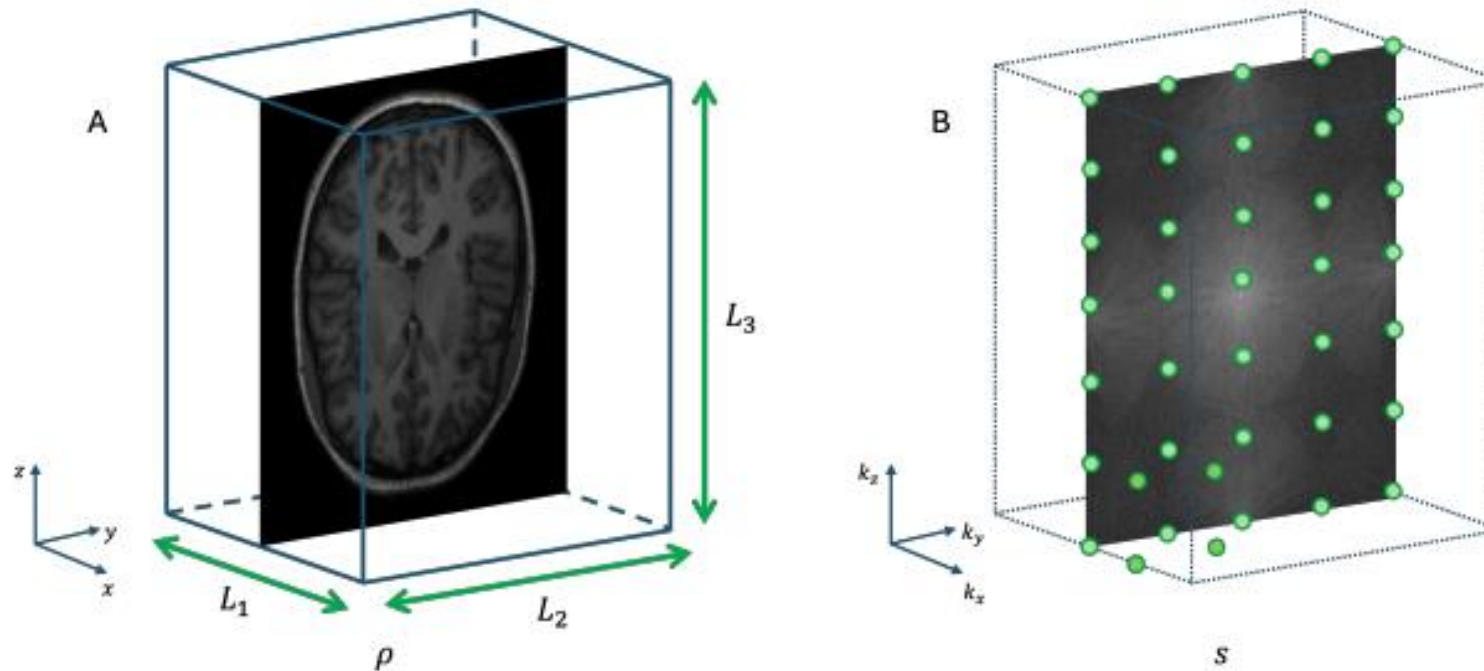
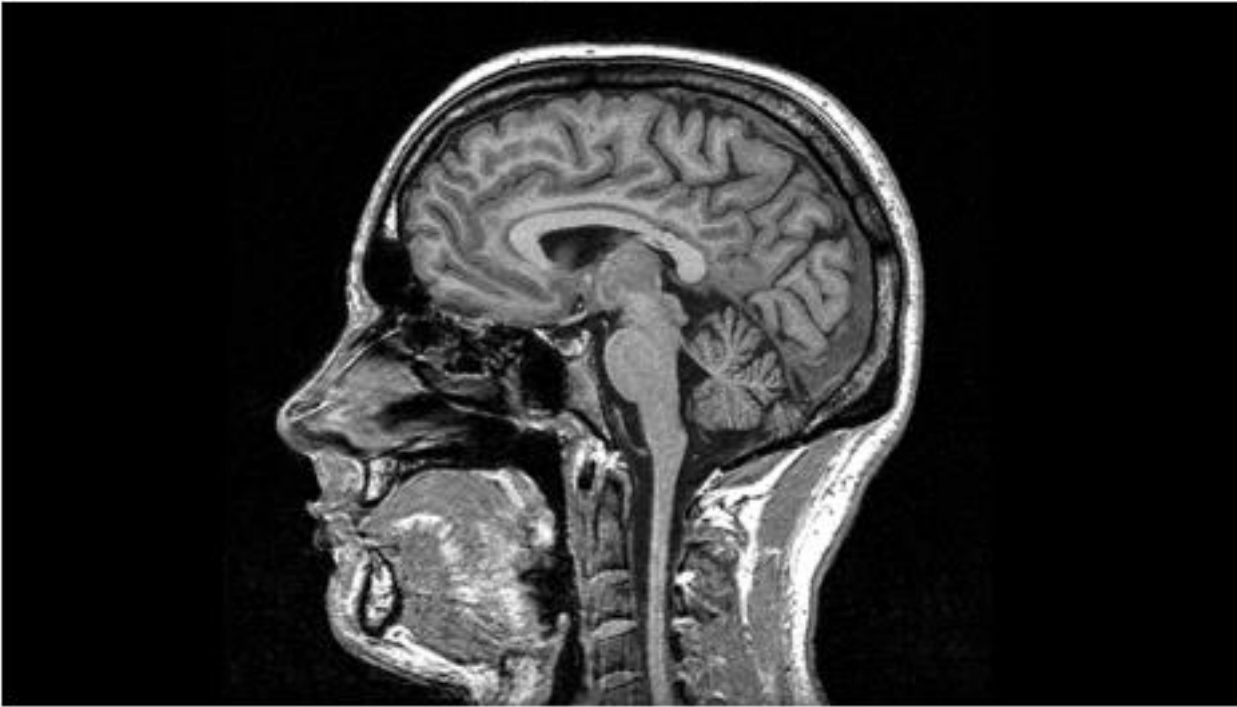


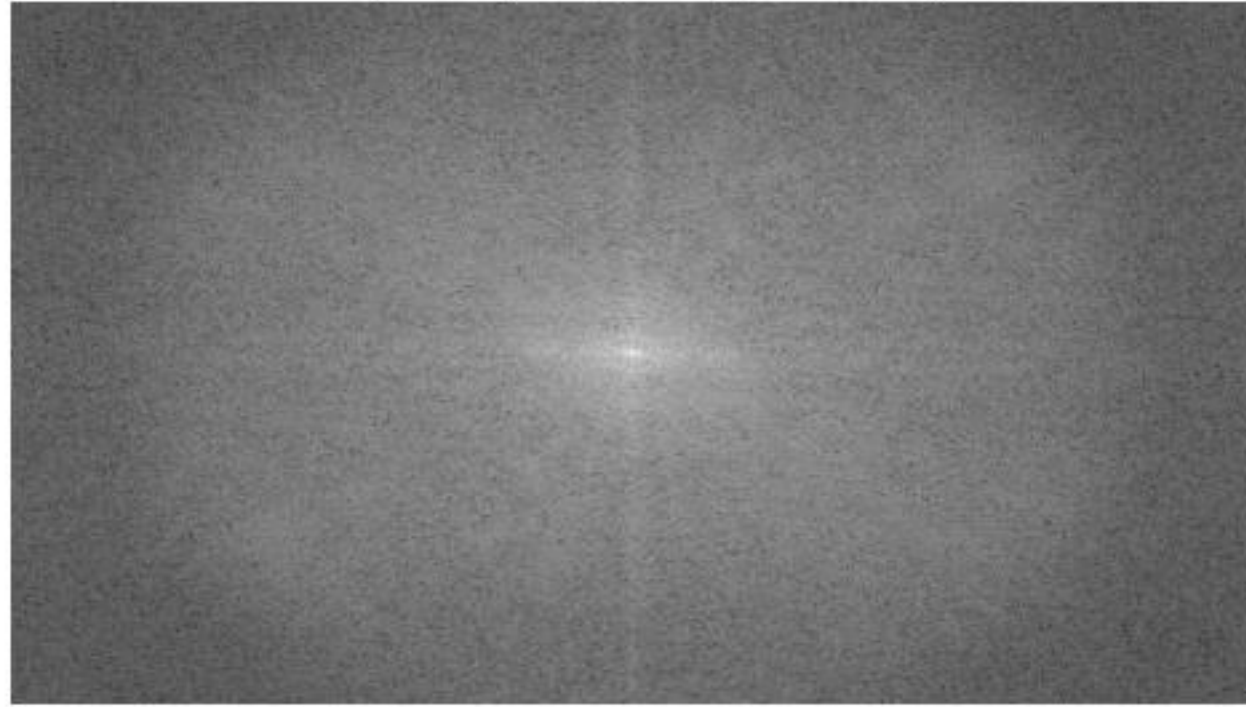
Figure 1.1: Full sampling of 3D MRI signal. (A) The proton density ρ is supported in a cube of sides L_1 , L_2 and L_3 of the image domain, i.e., the Fourier transform of s is supported in $[-L_1/2, L_1/2] \times [-L_2/2, L_2/2] \times [-L_3/2, L_3/2]$. (B) Every coordinate direction of the k-space is sampled at its Nyquist frequency.

Magnetic Resonance Imaging

Original Image



K-space (Frequency Domain)



2. EYE-TRACKER

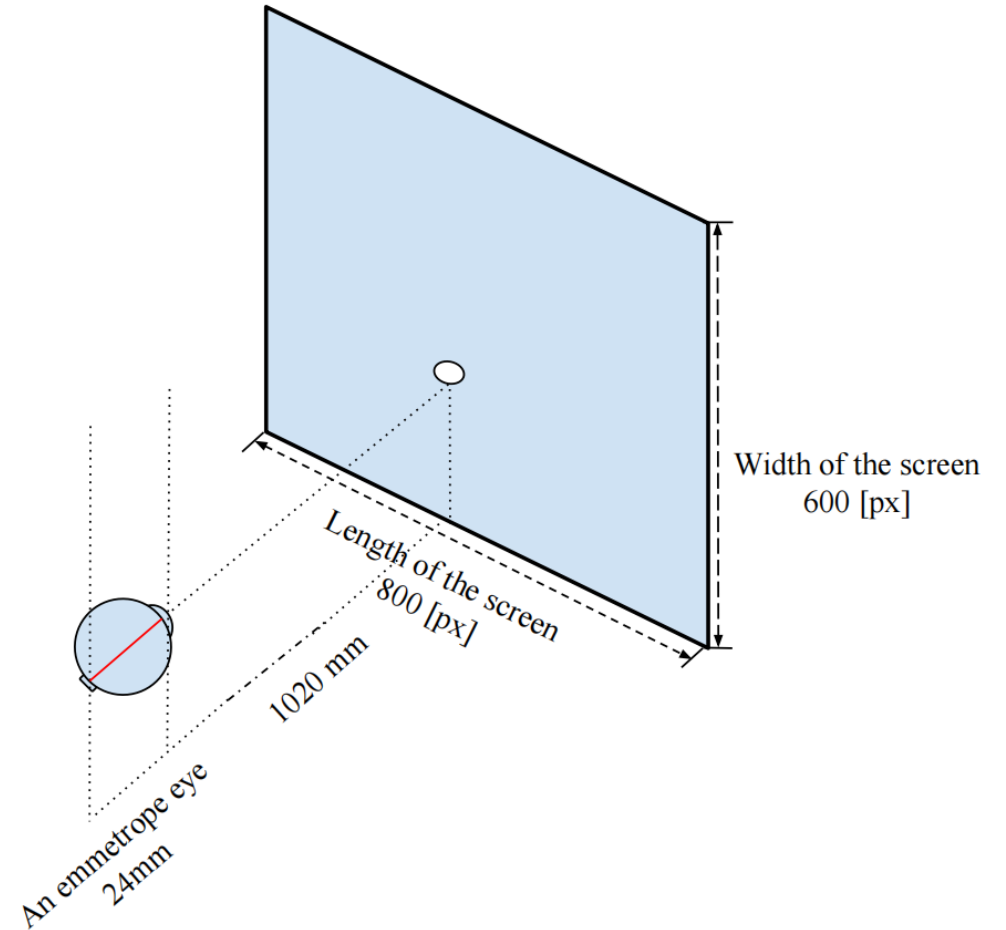
Eye-tracker

Eye tracking

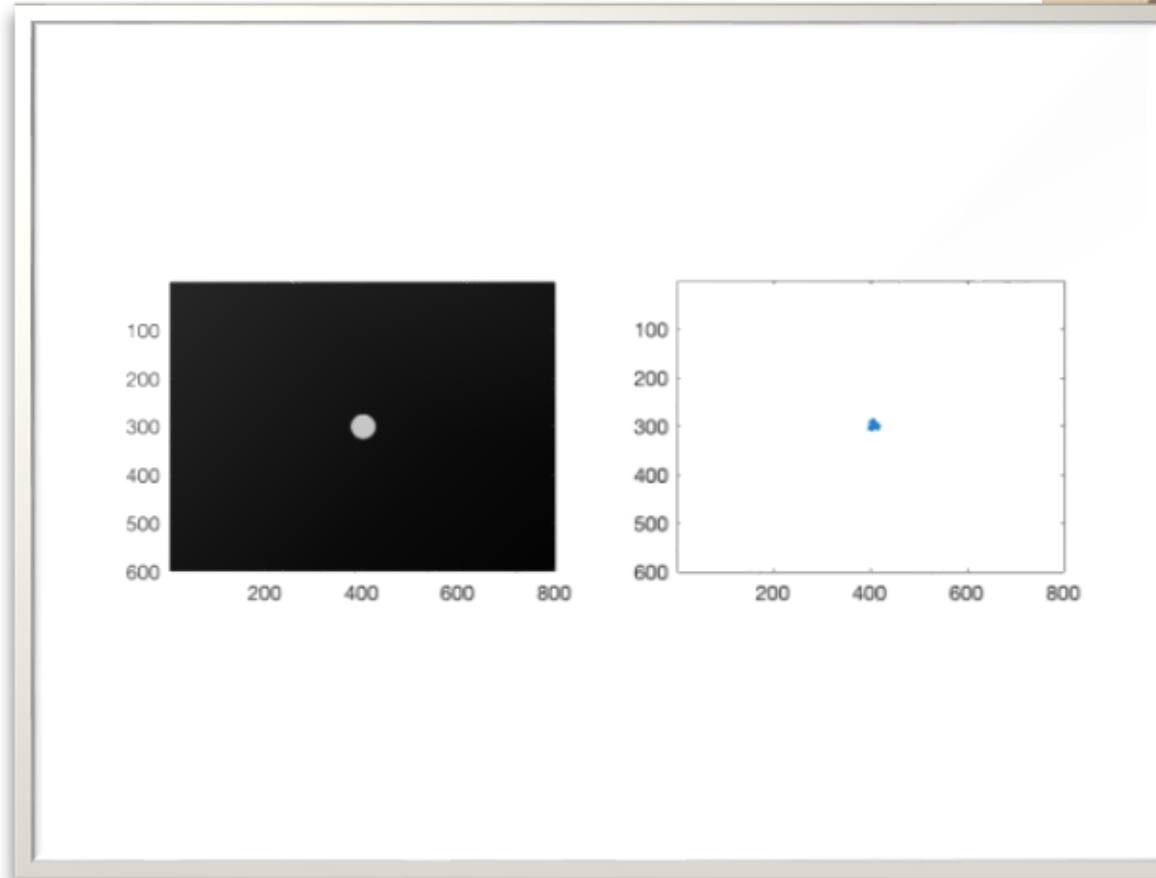


Eye Tracker (EyeLink 1000 Plus)

Eye-movements will be recorded via the Eye-Tracking installed within the MRI room and synchronized with the MR acquisition.



Eye-tracker



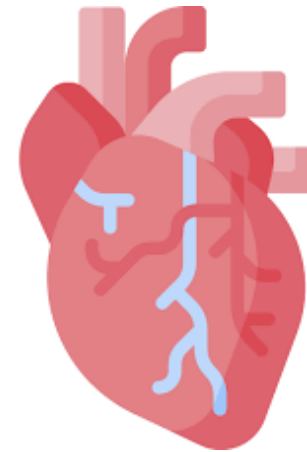
2. ELECTRO-CARDIOGRAM (ECG)

Electro-Cardiogram (ECG)

An **electrocardiogram (ECG or EKG)** is a quick test to check the heartbeat. It records the electrical signals in the heart. Test results can help diagnose heart attacks and irregular heartbeats, called arrhythmias. (Mayo Clinics)

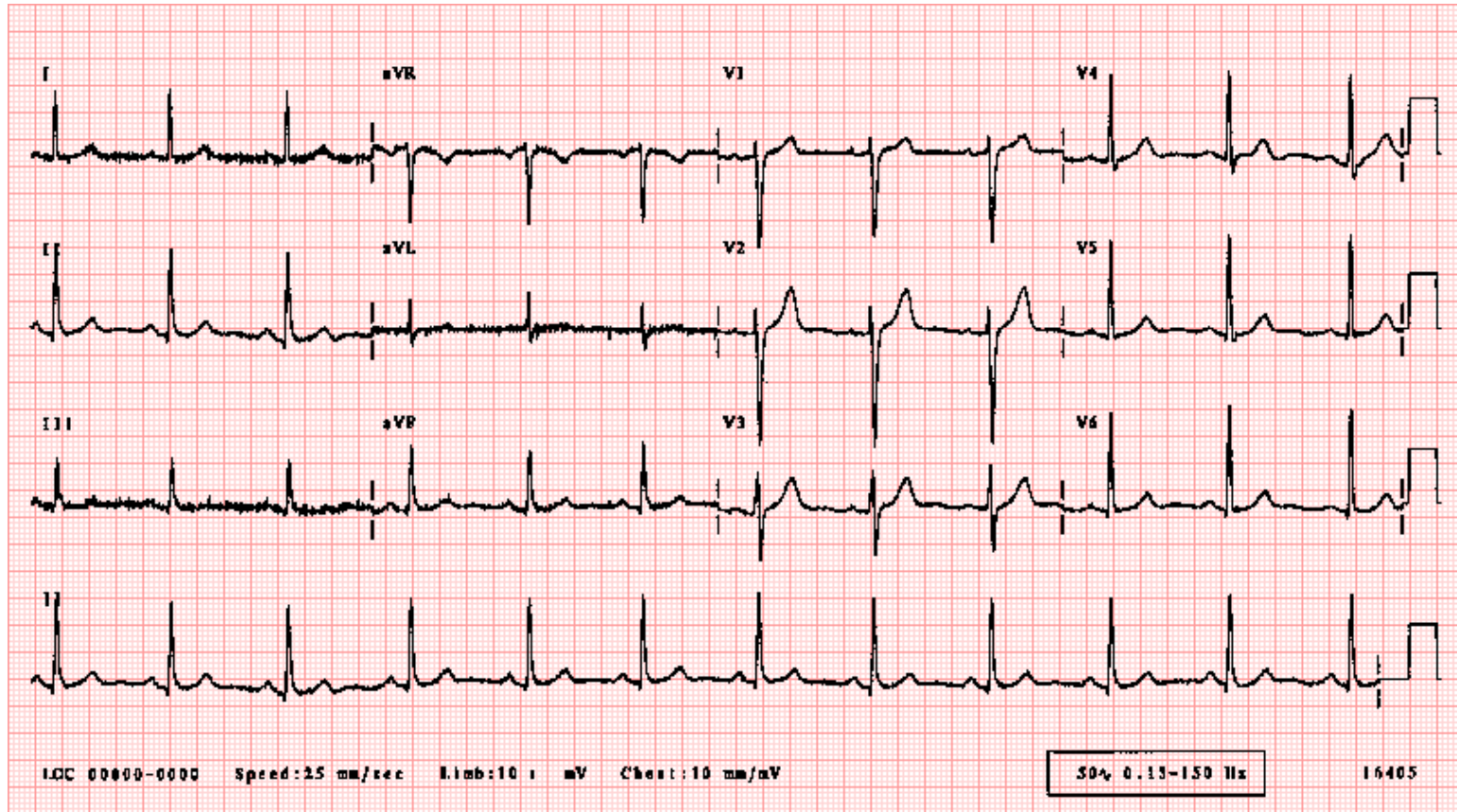
Among other condition, we can disclose also other causes of chest pain, or a previous heart attack.

Mathematically, an ECG is a line graph of the heart's electrical activity through repeated cardiac cycles.

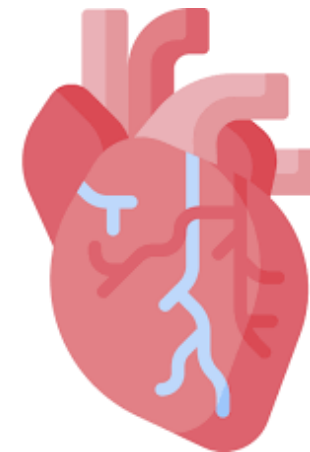
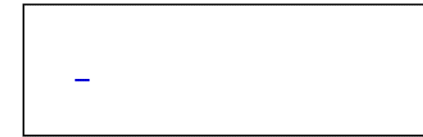
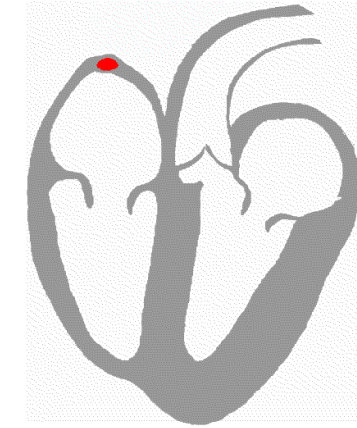


Electro-Cardiogram (ECG)

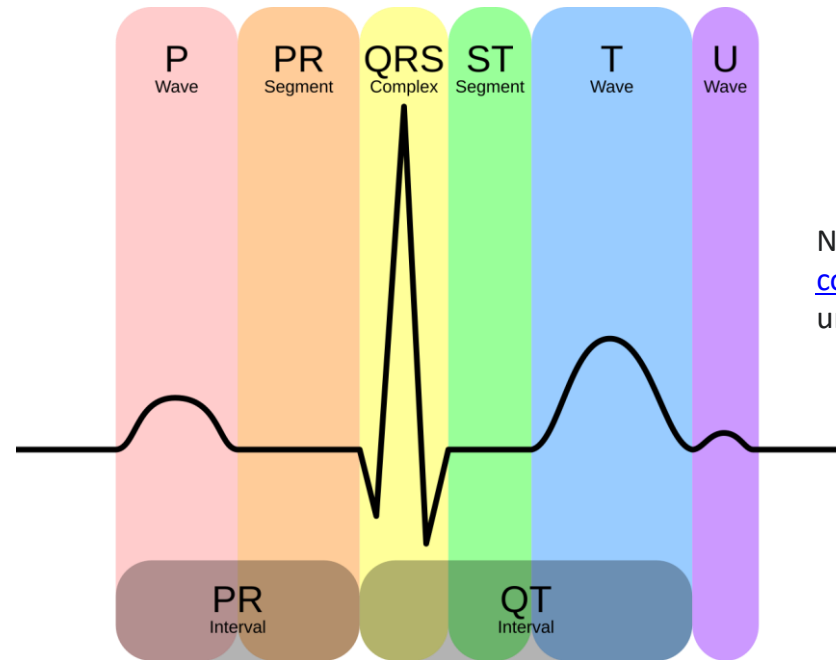
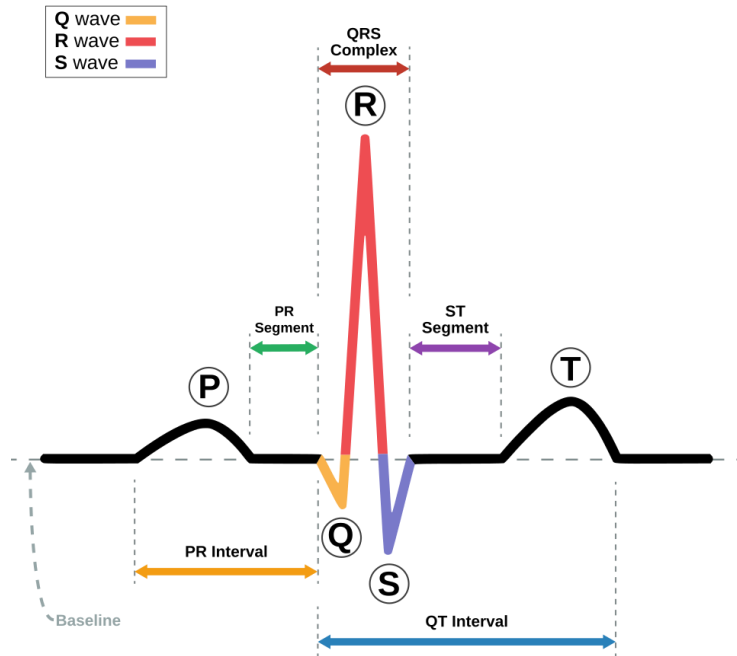
A normal ECG



Source: <http://www.ecglibrary.com/ecghome.php>
and By Kalumet - selbst erstellt = Own work, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=438140>



Electro-Cardiogram (ECG)



Normal rhythm produces four entities – a [P wave](#), a [QRS complex](#), a [T wave](#), and a [U wave](#) – that each have a fairly unique pattern:

- The P wave represents atrial depolarization.
- The QRS complex represents ventricular depolarization.
- The T wave represents ventricular repolarization.
- The U wave represents papillary muscle repolarization.

Created by [Agateller \(Anthony Atkielski\)](#)

Electro-Cardiogram (ECG): physics of electrodes

Electrodes are the actual conductive pads attached to the body surface. Any pair of electrodes can measure the [electrical potential difference](#) between the two corresponding locations of attachment.

ECG measures the heart's electrical activity (depolarization / repolarization) by detecting minute potential differences (< 5 mV) on the skin's surface, acting as a volume conductor. Electrodes convert ionic currents in the body to electronic currents via a conductive gel, while amplifiers filter noise to produce 12-lead, time-voltage, tracings.

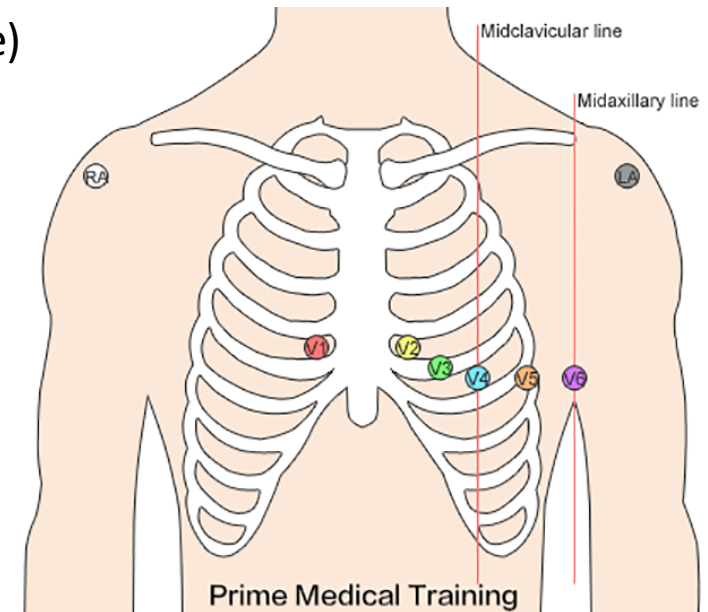
Among many sources:

- <https://derangedphysiology.com/main/cicm-primary-exam/cardiovascular-system/Chapter-735/physiological-basis-electrocardiography>
- <https://www.robots.ox.ac.uk/~gari/teaching/cdt/A3/readings/ECG/ch1.pdf>



Electro-Cardiogram (ECG): experimental setup

- The person should be placed always in the same position, suggestion: laying down
- **ALL experimental parameters** need to be controlled, meaning: are the eyes closed? Did he/she cough during the recordings? Major movements happen? (etc)
- If yes, how all these parameters translate in the recordings?
- A consistent configuration for electrodes placement needs to be found (in literature)
Example: <https://litfl.com/ecg-lead-positioning/>





Lesson 6 – part 2. How to treat a signal from A-Z

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1. UNDERSTAND YOUR DATA (DO YOU HAVE INFORMATION ABOUT THE SOURCE?)

Investigate the source of your data



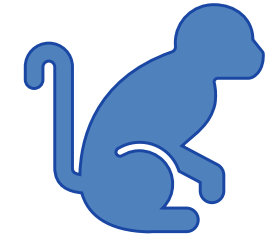
Questions:

- What is the **Channel** (human, machine, plant, animal)?
- Where these data have been acquired? – Understand the **environment** (gives hints about the preprocessing)
 - What is the source of **noise**?
 - How can I **filter** this noise?
- What is the **dimensionality** of the signal?
- What is the **sampling rate**?
- What is my **sample size**? Aka: how many individuals am I recording?

Which process am I investigating?

- Am I trying to make prediction?
- Am I trying to build a model of behaviour?

Use case 1: Vervet monkeys

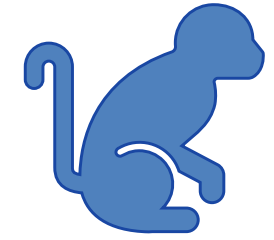


Questions:

- Channel:
- Environment:
 - Noise:
 - How can I filter this noise:
- Dimensionality of the signal:
- What is the sampling rate?
- Sample size:

Which process am I investigating?

Use case 1: Vervet monkeys



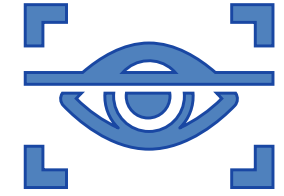
Questions:

- Channel: Monkeys via collar microphones
- Environment: South Africa wild reserve
 - Noise: Wild sounds and other monkeys
 - How can I filter this noise: Peak identification, high-pass filter
- Dimensionality of the signal: 1D
- What is the sampling rate? Microphones specifics
- Sample size: 10-15

Which process am I investigating?

- Analyze peaks, correspondences, differences of the signals across monkeys

Use case 2: Eye-Tracker



Questions:

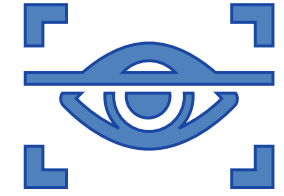
- Channel: Human eye-movements via infra-red camera
- Environment: Dark room, possible head movements
 - Noise: Blinks, Eye-movements, Sampling failure (device error)
 - How can I filter this noise: Erase data points affected by artefacts
- Dimensionality of the signal: 1D and 2D (why?)
- What is the sampling rate? Infra-red camera specifics
- Sample size: 4

Which process am I investigating?

- Gaze stability (fixation ability of the subjects), Deviation from center



Use case 3: Eye-Tracker 2.0



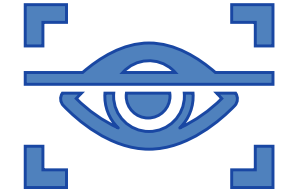
Questions:

- Channel: Human eye-movements via infra-red camera
- Environment: Dark room, possible head movements
 - Noise (artefacts):
 - How can I filter this noise:
- Dimensionality of the signal:
- What is the sampling rate?
- Sample size:

Which process am I investigating?

- What are the characteristics of healthy versus patients in their gaze trajectories?

Use case 3: Eye-Tracker 2.0



Questions:

- Channel: Human eye-movements via infra-red camera
- Environment: Dark room, possible head movements
 - Noise (artefacts): Blinks, Eye-movements out of range, Sampling failure (device error), patient's disease
 - How can I filter this noise: Erase data points affected by artefacts
- Dimensionality of the signal: 1D and 2D
- What is the sampling rate? Infra-red camera specifics
- Sample size: 10-20

Which process am I investigating?

- What are the characteristics of healthy versus patients in their gaze trajectories?

Code: https://github.com/bfranceschiello/EyeTracking_preprocessing_and_ML_analysis

Data: <https://zenodo.org/records/6424677#.YsQ5suxBz0o>

Use case 4: Optical illusions

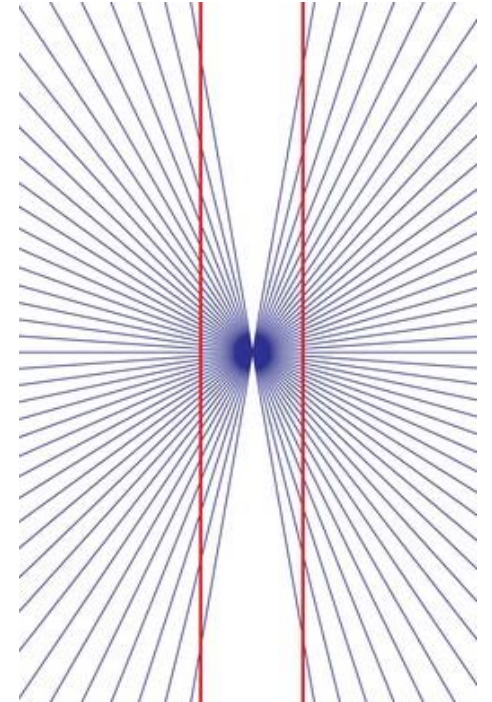
Questions:

- Channel: Grey-scale digital Images (0-1)
- Environment: Computer
 - Noise: No noise
 - How can I filter this noise: N/A
- Dimensionality of the signal: "2D" (but is 3D)
- What is the sampling rate? The signal does not evolve in time.
- Sample size: 5

Which process am I investigating?

- Calculate the energy of the image
- Extract orientation features

Article: <https://www.sciencedirect.com/science/article/pii/S2665945X24000172?via%3Dihub>



Use case 5: Natural Images

Questions:

- Channel: RGB digital Images
- Environment: Camera
 - Noise: Environment noise: light exposition
 - How can I filter this noise: Filtering
- Dimensionality of the signal: "2D" (2x3 tensor ;-)
- What is the sampling rate? The signal does not evolve in time
- Sample size: 10 (but on online database you can get as many as you want)

Which process am I investigating?

- Calculate the energy of the image
- Extract features



Take home message

Your **pre-processing** and **data preparation** strongly depend on what is the questions you're trying to answer to.

Important!!